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Research Assignment 2

1. Business Analytics

The process organizations use to evaluate and monitor their performance. The insights it help address current challenges and anticipate future issues.

Example: HR notice less overtime and more sick leave, leading to lower productivity. Using HR analytics, they found the company was understaffed and decided to hire more employees.

2. Business Intelligence

Involves collecting and transforming data into visualizations, reports and dashboards to uncover trends, patterns and key insights. These insights enable organizations to assess performance and make informed, data-driven decisions.

Example: Starbucks use business intelligence to analyze factors like demographics, location, and customer behavior to choose store locations and improve customer experience. It also tracks purchase data through loyalty programs and apps to predict buying patterns and send personalized offers, increasing sales.

3. Big Data

Refers to extremely large datasets that can be analyzed to uncover patterns, trends and relationships, especially in human behavior. It is important because it enables better decision-making, reveals new insights and opportunities, and allow for personalized experiences.

Example: Personalized recommendations, platforms like Amazon use Big Data to suggest products based on your browsing history, past purchases and user behavior.

These systems apply algorithms like collaborative filtering to identify patterns and predict what you are likely to buy.

4. Diagnostic analytics

Is a type of business analytics that examines historical data to identify root causes, patterns, and relationships. It uses methods like data mining, ~~correlation~~ correlation analysis, drill-down techniques, and statistical modelling.

Example: In health care use it to examine why certain patients have longer recovery times, identifying causes of high readmission rates, and determining which treatments are most effective for specific conditions.

5. Prescriptive Analytics

Uses data ~~recommend~~ to recommend the best possible actions to take. It considers various factors to suggest optimal next steps for decision-making. This makes it a powerful tool for supporting data-driven decisions.

Example: Sales lead scoring, ~~pres~~ prescriptive analytics help rank leads based on their likelihood to convert by assigning scores to actions like page views, email clicks, and content engagement. Higher scores are given to actions showing purchase intent, helping sales teams focus on the most promising leads and improve efficiency.

6. Data Analytics

Is a multidisciplinary field that uses mathematics, statistics, and computer science to extract insights from data. It covers everything from analyzing existing data to designing methods for collecting and storing it.

Example: A sneaker manufacturer can use sales data to decide which designs to keep and which to discontinue.

7. Data-driven decision making

Is the practice of using data and analysis rather than intuition to guide business choices. It relies on sources like customer feedback, market trends, and financial data to support better decisions.

Example: Amazon uses data to group customers based on location, demographics, and purchasing behavior. This helps them create targeted marketing campaigns.

and improve customer engagement

8. Descriptive Analytics

Identifies key patterns and trends in large datasets and is the most commonly used form of data analysis. It focuses on creating metrics and key performance indicators (KPIs) for dashboards and business reports.

Example: A hospital can analyze patient records to find patterns like average emergency room wait times, common diagnoses, and admission trends over time. By using summaries and visuals like charts, they can identify issues and improve areas such as staffing and patient flow.

9. Predictive Analytics

Uses historical data to forecast future trends and outcomes. It helps organizations anticipate what may happen next and support better strategic decision-making.

Example: Finance (cash flow forecasting) predictive analytics use past ~~financial~~ financial and industry data to forecast future sales, and expenses. This helps business plan ahead and make informed financial decisions despite uncertainty.

10. Data Literacy

Data literacy is the ability to understand, analyze, visualize, and use data effectively to achieve goals. How it is applied depends on the specific context or situation.

Example: Beck's Hybrids improve its data literacy by training sales teams on how to use data and working closely with dealers to strengthen understanding. This led to better reporting and increased self-service use of data across the business.

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11 Structured Data

Is data that is organized in a fixed format, making it easy for both humans and machines to read. It is usually stored in rows and columns based on a defined structure or schema.

For Example: A customer database with fields like name, address, phone number, and email is structured.

12. Unstructured Data

Is information that doesn't follow a fixed format or structure. It is often disorganized, messy, and difficult to organize or analyze.

Example: Social media content such as posts, tweets, comments, and images on platforms like Facebook, Instagram and Twitter.

13. Data Warehouse

Is a central system where businesses store important data like customer and sales information for analysis and reporting. It combines current and historical data from multiple sources through ETL processes to support business intelligence and decision-making.

Example: A banking statement contains many transactions such as withdrawals, purchases and payments. With millions of customers, banks generate huge amounts of this data daily. They use data warehouses to store and analyze it in order to gain insights and improve their services.

14 Data Mining

Is the process of analyzing large datasets to discover patterns, relationships, and trends. It combines statistics, database systems, and machine learning to uncover deeper insights beyond basic observations.

Example: In finance, it is used to monitor transactions, assess credit risk, and analyze portfolios. It can also detect unusual activity and support better lending and compliance decisions.

15 Data Visualization

Is the process of presenting data in visual formats such as charts, graphs, and maps to make it easier to understand. It helps people quickly interpret complex data and make informed decisions.

Example: during COVID-19, dashboards like the Johns Hopkins University COVID-19 dashboard, used maps and charts to show infection rates, recoveries, and deaths in real time. This helped governments and the public track the virus and make informed decisions quickly.

16 Dashboards

Is a visual tool that displays key data, metrics, and trends in one place. It provides a quick overview of performance to help users make faster and better decisions.

Example: Financial dashboards track KPIs like revenue, expenses, and profit to show the overall financial health of a company.

17 Metric

Is a measurable value used to track and assess performance. It helps businesses understand progress and guide decision-making.

Example: E-commerce companies use metrics like average order value (AOV), cart abandonment rate, and customer lifetime value (LTV) to evaluate performance.

18. Key Performance Indicator (KPI)

Are measurable values used to track progress toward specific business goals. They help organizations evaluate performance and guide decision-making.

Example: Website traffic is a KPI that shows how many people visit a company's website and helps assess online engagement and potential sales.

19. Benchmark

Is the process of comparing a company's performance with others in the same industry to identify strengths and weaknesses, help business understand their competitive position and improve performance.

Example: Financial benchmarking compares revenue, costs, and overall financial performance with other companies, often using data from public firms.

20. Segmentation

is the process of dividing a large group into smaller groups based on shared characteristics. In business, it helps companies target customers more effectively through personalized marketing & service.

Example: segmented marketing campaigns often achieve higher engagement and conversion rates than general campaigns because they are tailored to specific groups.

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21. Conversion Rate

Is the percentage of users who complete a desired action out of the total number of users. It shows how effectively a website or app turns visitors into customers or leads.

Example: E-commerce, conversion include actions like signing up, adding items to a cart or completing a purchase, helping business identify where customers drop off.

22. Customer Acquisition Cost (CAC)

Is the total cost a business incurs to gain a new customer. It includes expenses like marketing salaries, and campaigns, and is often compared with customer lifetime value (LTV) to measure profitability.

Example: Marketing manager like Tim would calculate CAC by considering all campaign costs and resources used to acquire new customers over a period.

23. Customer Life time Value (CLV/LTV)

Is the estimated total profit a business expects to earn from a customer over the entire relationship. It helps companies understand customer value and optimize marketing and spending.

Example: Marketers use CLTV to measure campaign ROI and segment customers for more targeted and effective messaging.

24. Churn Rate

Is the percentage of customers who stop using a company's product or service over a specific period. It helps business understand customer retention and identify potential issues.

Example: If a company loses 120 out of 1000 customers in a quarter, its churn rate is 12% which can be used to track performance and compare with competitors.

25. Customer Retention Rate

Is the percentage of customers who continue using a product or service over a given period. A high retention indicates strong customer loyalty, while a low rate may

Signal underlying issues.

Example: Netflix improve retention by ~~introducing~~ introducing a lower-cost, ad-supported ~~plan~~ plan to re-engage users and stay competitive.

26. Net promoter Score (NPS)

Measures customer loyalty by asking how likely customers are to recommend a product or service on a scale of 0-10. It is calculated by subtracting the percentage of detractors (0-6) from promoters (9-10) giving a score between -100 and 100.

Example: Business use NPS to identify unhappy customers for immediate support, engage neutral customers, and leverage satisfied customer for referrals and growth opportunities.

27. Average Order Value (AOV)

Measures the average amount a customer spends per order, calculated by dividing total revenue by the number of orders. It helps business understand spending behaviour and improve sale strategies.

Example: A store makes 24 000 from 400 orders, the AOV is \$60, which can guide upselling, cross-selling, and promotional decisions.

28. Sale funnel

Is the process a customer goes through from first learning about a product to making a purchase. It helps business track progress and guide prospects towards buying.

Example: A store can attract customers (awareness), build ~~that~~ interest, address concerns, and offer incentives to convert them into buyers.

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29. Marketing Campaign

is the promotion of a product or service using different media channels such as TV, radio, print and online platforms.

Example: Burger King used a viral marketing campaign by targeting a competitor's decision to discontinue a popular item and promoting a similar product to attract customers.

30. Post-campaign Analysis

is the evaluation of a marketing campaign after it ends by reviewing data and KPIs to assess performance and improve future strategies.

Example: Tools like Camhouse help marketers analyze results by centralizing data, providing real-time insights and supporting team collaboration for better decision-making.

31. Incremental Revenue

is the extra income generated from an additional sale or marketing effort compared to a baseline. It is used to measure the impact of different strategies on revenue growth.

Example: retargeting campaigns such as ~~email~~ reminder emails for abandoned carts - encourage customers to return and complete purchases, increasing a company's overall revenue.

32. Return on Investment

is a metric used to measure how profitable an investment is. It compares the profit gained to the cost of the investment, helping evaluate performance and compare options.

Example: An investor earns a 20% ROI from bakery investment and a 40% ROI from a shoe business, showing that the shoe ~~bus~~ business is the more profitable option.

33. Return on Ad Spend

Measures how much revenue is generated for every unit of money spent on advertising. It helps business evaluate ad performance, manage budgets, and improve campaign efficiency.

Example: financial professionals use ROAs to assess advertising effectiveness during

credit analysis, investment and performance forecasting.

34. Click-Through Rate

Is a metric that measures the percentage of people who click on an ad or link after seeing it. It shows how effective content is at attracting user engagement.

Example: Google Ads, a higher CTR means ~~that~~ the ad is relevant and appealing, which can improve its ranking and reduce advertising costs.

35. Bounce Rate

Is the percentage of visitors who leave a website after viewing only one page. It helps measure how engaging a page is and whether it encourages further interaction.

Example: If 45 out of 100 visitors leave after viewing just one page, the bounce rate is 45%, meaning nearly half of users did not continue browsing.

36. A/B Testing

Is a method where marketers compare two or more versions of a campaign element to see which performs better. It is used to improve results.

Example: Landing page can be tested by changing elements like headlines, images, or call-to-action button to see which version generates more conversions.

37. Lead Generation

Is the process of attracting and capturing interest in a product or service to convert potential customers into sales.

Example: Inforchare Academy generated leads by offering a free e-book through social media ads, using simple sign-up forms and automated emails to engage and convert users.

38. Market Attribution

Is the process of identifying which marketing channels and touchpoints influence a customer's decision to purchase. It helps businesses understand how different interactions contribute to conversions.

Example: in first-touch attribution, full credit for a sale is given to the first interaction a customer has with a brand, such as a linkedin ad that initially introduces them to a product.

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